

Worksheet — Algebraic Expressions and Polynomials (Class 10 NIOS Mathematics)

Total marks: **30** · Questions: **10**

Instructions: Attempt all 10 questions in your notebook. Show every step of your working for the subjective questions; choose the single best option for MCQs. Answers without working will not receive full marks.

Total time: 50 minutes.

Q1. How many terms does the expression $5p^3 - 2p^2 + p - 9$ have? [1 mark]

- (A) 2
- (B) 3
- (C) 4
- (D) 5

Q2. The expression $\frac{3}{x} + 2$ is a polynomial. [1 mark]

- (A) True
- (B) False

Q3. In the term $-7x^2y$, the coefficient is _____. [1 mark]

Q4. What is the degree of the polynomial $4xy + 7x^2y^3 - 2$? [2 marks]

- (A) 2
- (B) 3
- (C) 4
- (D) 5

Q5. Which of the following is the standard form of $z^2 - 1 + 5z^4$? [1 mark]

- (A) $z^2 + 5z^4 - 1$
- (B) $5z^4 + z^2 - 1$
- (C) $-1 + z^2 + 5z^4$
- (D) $5z^4 - 1 + z^2$

Q6. Add $(3x^2 - 5x + 2)$ and $(x^2 + 7x - 6)$ using the horizontal method. Show every step. [3 marks]

Q7. Subtract $(x^2 + 7x - 6)$ from $(3x^2 - 5x + 2)$. Show all steps. [3 marks]

Q8. Use FOIL to expand $(x + 5)(x - 2)$ and write the result in standard form. [4 marks]

Q9. Multiply $(2x^2 - 3x + 1)$ by $(x - 4)$ using the vertical (long-multiplication) method. State the degree of the product and explain how it is related to the degrees of the factors. [7 marks]

Q10. Let $p(x) = x^2 - 4x + 4$. Compute $p(2)$ and $p(3)$, and decide which (if any) is a zero of p . [7 marks]

Answer Key

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Q1. How many terms does the expression $5p^3 - 2p^2 + p - 9$ have?

Answer: 4

Terms are separated by + or – signs. The expression has four parts: $5p^3$, $-2p^2$, p and -9 .

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Q2. The expression $3/x + 2$ is a polynomial.

Answer: False

A polynomial cannot have a variable in the denominator. $3/x = 3 \cdot x^{-1}$, a negative power, so the expression is not a polynomial.

TRUE_FALSE · EASY · 10-NIOS-MATH-AEP-04

Q3. In the term $-7x^2 y$, the coefficient is ____.

Answer: -7

The coefficient is the numerical part of the term including its sign.

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Q4. What is the degree of the polynomial $4xy + 7x^2 y^3 - 2$?

Answer: 5

Degree of each term is the sum of exponents of all variables. $4xy$ has degree $1+1=2$; $7x^2 y^3$ has degree $2+3=5$; the constant has degree 0. The polynomial degree is the maximum, 5.

MCQ · MEDIUM · 10-NIOS-MATH-AEP-04

Q5. Which of the following is the standard form of $z^2 - 1 + 5z^4$?

Answer: $5z^4 + z^2 - 1$

Standard form lists terms in descending order of degree: degree 4, then degree 2, then the constant.

MCQ · EASY · 10-NIOS-MATH-AEP-04

Q6. Add $(3x^2 - 5x + 2)$ and $(x^2 + 7x - 6)$ using the horizontal method. Show every step.

Answer: $(3x^2 - 5x + 2) + (x^2 + 7x - 6) = (3x^2 + x^2) + (-5x + 7x) + (2 - 6) = 4x^2 + 2x - 4$.

Group like terms: x^2 terms $3x^2 + x^2 = 4x^2$; x terms $-5x + 7x = 2x$; constants $2 + (-6) = -4$. Sum = $4x^2 + 2x - 4$.

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Q7. Subtract $(x^2 + 7x - 6)$ from $(3x^2 - 5x + 2)$. Show all steps.

Answer: $(3x^2 - 5x + 2) - (x^2 + 7x - 6) = (3x^2 - 5x + 2) + (-x^2 - 7x + 6) = 2x^2 - 12x + 8$.

Flip the sign of every term being subtracted, then add: $3x^2 - x^2 = 2x^2$; $-5x - 7x = -12x$; $2 + 6 = 8$. Result = $2x^2 - 12x + 8$.

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Q8. Use FOIL to expand $(x + 5)(x - 2)$ and write the result in standard form.

Answer: $(x + 5)(x - 2) = x \cdot x + x \cdot (-2) + 5 \cdot x + 5 \cdot (-2) = x^2 - 2x + 5x - 10 = x^2 + 3x - 10$.

First: $x \cdot x = x^2$. Outer: $x \cdot (-2) = -2x$. Inner: $5 \cdot x = 5x$. Last: $5 \cdot (-2) = -10$. Combine like terms: $-2x + 5x = 3x$. Result = $x^2 + 3x - 10$.

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Q9. Multiply $(2x^2 - 3x + 1)$ by $(x - 4)$ using the vertical (long-multiplication) method. State the degree of the product and explain how it is related to the degrees of the factors.

Answer: Vertical layout — multiply $(2x^2 - 3x + 1)$ by x to get $2x^3 - 3x^2 + x$; multiply by -4 to get $-8x^2 + 12x - 4$; add column by column to get $2x^3 + (-3x^2 - 8x^2) + (x + 12x) + (-4) = 2x^3 - 11x^2 + 13x - 4$. Degree of product = 3 = $\deg(2x^2 - 3x + 1) + \deg(x - 4) = 2 + 1$.

Distribute every term of the second factor across every term of the first; combine like terms. The degree-3 product equals the sum of the degrees (2 and 1), which is the multiplication-of-degrees rule.

LONG_ANSWER · HARD · 10-NIOS-MATH-AEP-07

Q10. Let $p(x) = x^2 - 4x + 4$. Compute $p(2)$ and $p(3)$, and decide which (if any) is a zero of p .

Answer: $p(2) = 2^2 - 4 \cdot 2 + 4 = 4 - 8 + 4 = 0$, so $x = 2$ IS a zero of p . $p(3) = 3^2 - 4 \cdot 3 + 4 = 9 - 12 + 4 = 1$, so $x = 3$ is NOT a zero of p .

A zero of $p(x)$ is an input that makes the polynomial evaluate to 0. Here only $x = 2$ satisfies that.

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